

CIA Examination – Power BI

Course: M.Com | Marks: 30 | Duration: 1 Hour

Problem Statement

BrightPath EdTech is an educational analytics company that partners with schools to help them understand student performance patterns. The company collects data on students' study habits, attendance, lifestyle factors, family background, and academic outcomes.

The **Academic Analytics Head** at BrightPath wants to build an interactive **Student Performance Dashboard** to help school administrators and teachers identify factors that may be influencing student outcomes and to monitor overall academic performance.

As a **Data Analyst Intern** at BrightPath EdTech, you have been given the dataset (Dataset-2-student_performance_dataset.csv) containing details such as Student ID, Gender, Study Time (hours), Attendance Percentage, Sleep Hours, Parental Education, Internet Access, Extracurricular Activities, Part-Time Job status, Previous Grade, Final Exam Score, and Final Grade.

You are required to **import this dataset into Power BI Desktop and build a single-page interactive dashboard**, using suitable visuals of your choice, to address the following requirements.

Tasks

1. Import the given dataset (Dataset-2-student_performance_dataset.csv) into Power BI Desktop.
2. **Class Snapshot**
Display the Total number of Students, Average Attendance Percentage, and Average Final Exam Score.
3. **Grade Distribution**
Show the number of Students falling under each Final Grade category.
4. **Gender-wise Performance**
Show the Average Final Exam Score by Gender.
5. **Study Time vs Performance**
Show the relationship between Study Time (hours) and Final Exam Score for all students.
6. **Lifestyle Factors**
Show the relationship between Sleep Hours and Final Exam Score for all students.
7. **Family Background Analysis**
Show the Average Final Exam Score based on Parental Education level.

8. **Dashboard Presentation**

Arrange all visuals into a single/multiple pages, well-formatted dashboard page with a suitable title, consistent formatting, and neat alignment.

9. **Insight & Interpretation**

Based on your dashboard, write **one observation** about a factor that appears to influence student performance (e.g., attendance, study time, or sleep).

Total: 30 Marks

Notes for Candidates

- Choose the visual (bar chart, pie chart, map, card, etc.) that best represents each requirement.
- Visuals should be clearly labeled with appropriate titles and legends.
- Save the file as YourName_RollNo_StudentDashboard.pbix before submission.
- Upload files or zipped files in the classroom at the end of the test (.pbix and screenshots)
- After uploading don't do any kind of updates, else zero marks will be allocated